

Manager Summary

Clinical Ground-Truth Review of CATE Mortality Results

Scope and interpretation

The CSV results were reviewed as model-estimated changes in mortality risk for ICU latent clinical-state tags. A value of +0.10 is read as an estimated 10-percentage-point increase in mortality for patients with that proxy state. The variables are labeled as treatments in the modeling output, but most represent clinical conditions such as organ dysfunction, shock, inflammation, or severity rather than therapies.

Bottom line: The results are directionally useful but not uniformly reliable as causal effect sizes. The strongest positive signals largely match established ICU risk patterns; the shock and PhysioNet heme/coag findings conflict with clinical expectations and should be treated as weak or potentially misleading signals rather than ground truth.

Key result pattern

Highest positive estimates were concentrated in organ dysfunction and overall severity. MIMIC ranked HepaticDysfunction (+15.5 pp) and Severity (+10.0 pp) highest. PhysioNet ranked RenalFail (+12.5 pp), HepFail (+10.1 pp), Severity (+9.2 pp), and Metab (+8.7 pp) highest. These patterns are broadly consistent with critical-care literature linking multi-organ failure, kidney injury, liver dysfunction, metabolic stress, and coagulation abnormalities to higher ICU mortality [1-8].

What is strong vs. concerning

What is good about the results	What is bad or uncertain about the results
The model recovers the expected high-level hierarchy: organ dysfunction and global severity are the main mortality drivers, not isolated vital-sign abnormalities.	Shock is implausibly small in MIMIC (+2.2 pp) and negative in PhysioNet (-0.6 pp). True shock is a high-risk critical-care syndrome, so these estimates likely understate the total effect [1,9].
The highest positive signals are clinically coherent: kidney dysfunction, liver dysfunction, metabolic derangement, coagulation dysfunction, and global severity are all recognized mortality-risk states [2-8].	PhysioNet HemeFail is negative (-1.8 pp), which conflicts with literature showing thrombocytopenia and coagulation abnormalities are associated with worse ICU outcomes [7].
Cross-dataset overlap supports credibility: severity, renal dysfunction, metabolic derangement, inflammation, and respiratory failure are positive in both summaries.	Several treated groups show lower observed mortality than the full cohort even when mean_cate is positive. This means mean_cate should not be read as a simple observed risk difference.
The results are useful for management-level prioritization: the strongest findings point to the same broad risk domains emphasized in ICU severity scores such as SOFA [2].	The largest MIMIC signal, HepaticDysfunction, is plausible but may partly represent broader multiorgan failure rather than an isolated liver-specific mortality effect [4,5].
MIMIC generally appears more clinically interpretable than PhysioNet because it contains richer treatment and process evidence; PhysioNet relies more heavily on measurement-only proxies.	The labels are proxy states, not confirmed diagnoses. PhysioNet in particular lacks vasopressors, antibiotics, diagnoses, cultures, and procedures, limiting interpretation of shock, sepsis, and cardiac-related tags.

Manager interpretation

The results are credible as a ranking of broad clinical risk domains. They are not credible as final numeric estimates of how much each condition independently causes death. The most trustworthy message is that severe organ dysfunction and overall severity carry substantial mortality signal. The least trustworthy message is the low or negative apparent impact of shock and PhysioNet heme/coag dysfunction, which contradicts established clinical knowledge.

For non-medical stakeholders: the model is mostly identifying the right danger zones, but some effect sizes are distorted by proxy labels, overlapping clinical pathways, and differences between the two datasets.

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Appendix A — Numeric Mortality Values Reported in Cited Sources

The values below provide clinical context for interpreting the highest model estimates. They are not direct replacements for mean_cate: the papers report cohort mortality rates, syndrome-definition mortality, or attributable mortality under their own definitions, while the project variables are proxy latent-state labels.

Cited source	Numeric mortality value reported	Relevant project variable(s)	Interpretation for manager review
Sepsis-3, JAMA 2016 [1]	Sepsis with SOFA increase ≥ 2 : >10% in-hospital mortality; septic shock criteria: >40% hospital mortality.	Shock; Inflammation/Sepsis; Global Severity	Supports the direction that sepsis-like physiology and shock should be high-risk states. Very small or negative model estimates for shock are not clinically credible as total mortality effects.
Berlin ARDS Definition, JAMA 2012 [8]	ARDS mortality increased with severity: 27% mild, 32% moderate, 45% severe.	Respiratory Failure	Supports a severity gradient for respiratory failure. The model should rank severe respiratory dysfunction as materially risky, but exact CATE size may differ from ARDS cohort mortality.
Yang et al., Frontiers in Medicine 2021 [4]	Bilirubin ≥ 2 mg/dL: 31.9% hospital mortality vs. 17.0% when bilirubin <2 mg/dL; after matching: 32.2% vs. 24.8%.	Hepatic Dysfunction	Supports hepatic dysfunction as a plausible high-mortality signal. The magnitude is consistent with the model ranking liver dysfunction among the stronger positive effects.
van Beest et al., Annals of Intensive Care 2013 [6]	Lactate normalization <6 hours: 16.6% mortality vs. 24.4% when normalization took >6 hours.	Metabolic Derangement; Lactate Burden; Shock	Supports lactate/metabolic burden as clinically meaningful. It also shows that timing and persistence matter, so a single binary tag may simplify the true risk relationship.
Wang et al., BMC Nephrology 2024 [11]	Overall 30-day attributable mortality of AKI among critically ill septic patients: 6.6%; stage 3 AKI: 16.8%.	Renal Dysfunction	Supports renal dysfunction as a real contributor to mortality, especially severe AKI. It also suggests that mild kidney dysfunction may have a smaller independent effect than broad model estimates imply.

Management implication: the cited literature supports the broad ranking of high-risk domains, but it does not validate the exact numeric CATE values. The project results should therefore be interpreted as directional risk signals rather than definitive causal mortality increments.